

Topics in Error-Correcting Codes

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Lecture 6: Interactive Coding (Cont.) (September 28, 2026)

Warning: This document is a rough draft, so it may contain bugs. Please feel free to email me with corrections.

Recap

Recall that last class we defined the notion of a tree code and showed how to use a tree code to construct an interactive coding scheme.

Definition 1. A tree code, T , is an online code with,

$$T : \{0,1\}^* \rightarrow \Sigma^*,$$

such that the following conditions hold:

- $|x| = |y| \implies |T(x)| = |T(y)|$.
- For every $x \in \{0,1\}^*$, $T(x)$ is the prefix of $T(x\|0)$ and $T(x\|1)$.
- For every $t \geq 1$ and distinct $x, x' \in \{0,1\}^t$, letting s be the first position at which they differ,

$$\Delta(T(x), T(x')) \geq \delta c(t - s + 1),$$

where Δ denotes Hamming distance measured in bits.

- There exists a constant $c \in \mathbb{N}$ such that $|\Sigma| = 2^c$.

Remark. We identify Σ with $\{0,1\}^c$. Throughout these notes, Hamming distance, denoted by Δ , and Hamming weight, denoted by wt , count individual bits, rather than alphabet symbols in Σ . Thus, a tree code with relative distance δ means that two paths that first differ at position s have encodings at bit distance at least $\delta c(t - s + 1)$ at depth t .

Last class we showed how to use a tree code to construct an interactive coding scheme.

Theorem 2 (Tree codes imply interactive coding). [1] Suppose that T is a tree-code of rate $R = 1/c$ and relative distance $\delta > 0$, where c is a positive integer. Then there is a generic deterministic transformation taking any binary protocol communicating k bits into a protocol communicating $12k/R$ bits, such that both parties recover the original transcript even if an adversary corrupts a $\delta/96$ fraction of the transmitted bits.

Schulman's Interactive Coding Scheme. At a high-level, at the end of round t each party $P \in \{A, B\}$ computes a pair (a_P^t, v_P^t) , where $a_P^t \in \{0, 1, \text{Back}, \text{Stay}\}$ is the move that it makes at the end of round t , and v_P^t is a vertex in the protocol tree of the original protocol Π that party P believes they are at. Initialize

$$a_P^0 = \text{Stay} \quad v_P^0 = \varepsilon.$$

Party P also maintains a history $(h_P^1, \dots, h_P^t)$ where each

$$h_P^i \in \Sigma := \{0, 1, \text{Back}, \text{Stay}\} \times \{0, 1\},$$

where

$$h_P^t = (a_P^{t-1}, b_P(v_P^{t-1})),$$

where $b_P(v_P^{t-1})$ is the bit that P would send in protocol Π if they were in round v_P^{t-1} , and if this vertex is owned by the other party Q then we define $b_P(v_P^{t-1}) = 0$. Note that $h_P^{(t)} = (h_P^1, \dots, h_P^t)$ defines a vertex in the protocol tree of Π . This vertex, denoted by $\text{Vertex}(h_P^{(t)})$, is defined by the actions in this history vector,

$$(a_P^0, \dots, a_P^{t-1}).$$

Note that

$$\text{Vertex}(h_P^{(t)}) = v_P^{t-1}.$$

The interactive coding scheme consists of $N = 2k$ rounds, where k is the number of rounds in the underlying protocol. In each round both parties transmit a message simultaneously. In round t , party P , with the other party denoted by Q , performs the following steps:

1. Append h_P^t to its history, and send the t th output block of

$$\tilde{T}(h_P^1, \dots, h_P^t).$$

where

$$\tilde{T}(h_1, \dots, h_t) = T(e(h_1), \dots, e(h_t))$$

where $e : \{0, 1, \text{Stay}, \text{Back}\} \times \{0, 1\} \rightarrow \{0, 1\}^3$ is an arbitrary bijection.

Note that each block consists of $L = 3c$ bits, and $\tilde{T}$ has relative distance $\alpha = \delta/3$.

2. Compute (a_P^t, v_P^t) from the (possibly corrupted) history blocks that it received from the other party, as follows:

- (a) Let $z_Q^t \in \{0, 1\}^{L \cdot t}$ be the concatenation of all blocks received from Q through round t . Compute

$$\hat{h}_Q^{(t)} \in \arg \min_{h \in \Sigma^t} \Delta(\tilde{T}(h), z_Q^t),$$

breaking ties according to a fixed rule. Set

$$\hat{v}_Q = \text{vertex}(\hat{h}_Q^{(t)}), \quad \hat{b}_Q = \text{Bit}(\hat{h}_Q^{(t)}),$$

The value $\text{Bit}(\hat{h}_Q^{(t)})$ is the second component of the last symbol of the decoded history. Namely, if $\hat{h}_Q^t = (\hat{a}, \hat{b})$, then

$$\text{Bit}(\hat{h}_Q^{(t)}) = \hat{b}.$$

(b) Choose a_p^t as follows:

- If $v_p^{t-1} = \hat{v}_Q$, then set $a_p^t = b_p(v_p^{t-1})$ if P owns this vertex, and set $a_p^t = \hat{b}_Q$ otherwise.
- If v_p^{t-1} is a proper ancestor of $\hat{v}_Q$, set $a_p^t = \text{stay}$.
- Otherwise, set $a_p^t = \text{back}$.

Then update

$$v_p^t = \text{Move}(v_p^{t-1}, a_p^t).$$

Note that the **histories are append-only**: a move back changes the protocol-tree vertex but does not erase any history symbol.

After round N , each party outputs the first k bits of its vertex, or outputs failure if its vertex has depth less than k . Any padding is then removed.

Each round communicates $2L$ bits, so the total communication is

$$M = 2LN = 6cN = 12ck. \quad (1)$$

Thus, there is a $12c$ overhead in communication complexity; in other words, the rate of the interactive coding scheme is $\frac{1}{12c} = \frac{R}{12}$.

Correctness of the Schulman's Interactive Coding Scheme

We now prove correctness. Fix the inputs and an arbitrary pattern of channel corruptions. Let $\pi^* \in \{0, 1\}^N$ be the correct transcript of the extended protocol on these inputs.

For every $t \in \{0, \dots, N\}$, let w_t be the longest common prefix of v_A^t and v_B^t . Note that w_N is a prefix of π^* . This is the case since at an Alice-owned vertex, this edge is consistent with Alice's input, and at a Bob-owned vertex, it is consistent with Bob's input. We need to prove that at the end of the protocol $w_N \geq k$. We prove this via a potential function.

The potential function. For every $t \in [N]$, let $B_t \in \{0, 1\}$ indicate whether at least one party incorrectly decodes the other party's history in round t . Define

$$\Phi_t = 2|w_t| - \max\{|v_A^t|, |v_B^t|\} + 4 \sum_{s=1}^t B_s.$$

In particular, $\Phi_0 = 0$.

Claim 1. For every $t \in [N]$,

$$\Phi_t \geq \Phi_{t-1} + 1.$$

Claim 2.

$$\sum_{t=1}^N B_t \leq \frac{N}{8}.$$

We first note that Claim 1 together with Claim 2 implies the correctness of the simulation. To this end, first note that Claim 1, together with the fact that $\Phi_0 = 0$ implies that $\Phi_N \geq N$. Next observe that since w_N is a prefix of both v_A^N and v_B^N ,

$$\max\{|v_A^N|, |v_B^N|\} \geq |w_N|.$$

Consequently,

$$|w_N| \geq \Phi_N - 4 \sum_{t=1}^N B_t \geq N - 4 \sum_{t=1}^N B_t \geq N/2 \quad (2)$$

where the latter inequality follows from Claim 2. This together with the fact that $N/2 = k$ implies the correctness of the simulation assuming Claim 1 and Claim 2. It thus remains to prove these two claims.

Proof of Claim 1. Fix $t \in [N]$. We consider separately the cases $B_t = 0$ and $B_t = 1$.

Case 1: $B_t = 0$. Both parties correctly decode the other party's entire history through round t . There are three possibilities.

1. The vertices coincide: $v_A^{t-1} = v_B^{t-1} = v$. The owner of v follows its prescribed bit, and the other party follows the correctly decoded announcement of that bit. Both parties therefore move to the same child of v . Hence,

$$|w_t| = |w_{t-1}| + 1$$

and

$$\max\{|v_A^t|, |v_B^t|\} = \max\{|v_A^{t-1}|, |v_B^{t-1}|\} + 1.$$

2. One vertex is a proper ancestor of the other. The shallower party stays and the deeper party backs up. The longest common prefix is unchanged, and the maximum depth decreases by exactly 1. Thus,

$$|w_t| = |w_{t-1}|$$

and

$$\max\{|v_A^t|, |v_B^t|\} = \max\{|v_A^{t-1}|, |v_B^{t-1}|\} - 1.$$

3. The vertices are incomparable. Both parties back up. Each party either remains in its original branch below w_{t-1} or reaches w_{t-1} itself. Their longest common prefix is therefore unchanged, while both depths decrease by 1. Hence,

$$|w_t| = |w_{t-1}|$$

and

$$\max\{|v_A^t|, |v_B^t|\} = \max\{|v_A^{t-1}|, |v_B^{t-1}|\} - 1.$$

In all three possibilities, the term $2|w_t| - \max\{|v_A^t|, |v_B^t|\}$ increases by 1. Since $B_t = 0$, the cumulative number of bad rounds does not change. Consequently,

$$\Phi_t - \Phi_{t-1} = 1.$$

Case 2: $B_t = 1$. Regardless of the decoding outcome, each party changes its vertex by at most one edge. Therefore,

$$\max\{|v_A^t|, |v_B^t|\} \leq \max\{|v_A^{t-1}|, |v_B^{t-1}|\} + 1.$$

Also, each party preserves the first $\max\{|w_{t-1}| - 1, 0\}$ bits of its path. Hence,

$$|w_t| \geq |w_{t-1}| - 1.$$

Since the cumulative number of bad rounds increases by 1, we obtain

$$\begin{aligned} \Phi_t - \Phi_{t-1} &= 2(|w_t| - |w_{t-1}|) \\ &\quad - \left(\max\{|v_A^t|, |v_B^t|\} - \max\{|v_A^{t-1}|, |v_B^{t-1}|\} \right) + 4 \\ &\geq -2 - 1 + 4 \\ &= 1. \end{aligned}$$

This proves the claim. $\square$

Proof of Claim 2. Let C denote the total number of corrupted bits. We show that

$$\sum_{t=1}^N B_t \leq \frac{2C}{\alpha L}. \quad (3)$$

where C is the total number of corrupted bits, which is at most

$$C \leq \frac{\delta}{96} \cdot M = \frac{\delta}{96} \cdot 2LN.$$

Note that Equation (3) implies that

$$\sum_{t=1}^N B_t \leq \frac{2C}{\alpha L} \leq 2 \cdot \frac{\delta}{96} \cdot 2LN \cdot \frac{1}{\alpha L} = \frac{\delta}{96} \cdot 4N \cdot \frac{3}{\delta} = \frac{N}{8},$$

as desired.

We thus focus on proving Equation (3). To this end, fix a party $P \in \{A, B\}$. Let C_P denote the number of corrupted bits transmitted by P , and let

$$\mathcal{B}_P = \left\{ t \in [N] : \widehat{h}_P^{(t)} \neq h_P^{(t)} \right\}$$

be the set of rounds in which the other party incorrectly decodes P 's history. Here,

$$h_P^{(t)} = (h_P^1, \dots, h_P^t)$$

is P 's actual history through round t . For an interval $I \subseteq [N]$, let $C_P(I)$ denote the number of corrupted bits transmitted by P in rounds belonging to I .

Fix $t \in \mathcal{B}_P$. Let i be the first position at which $h_P^{(t)}$ and $\widehat{h}_P^{(t)}$ disagree, and write

$$s = \widetilde{T}(h_P^{(t)}), \quad \widehat{s} = \widetilde{T}(\widehat{h}_P^{(t)}), \quad z = z_P^t.$$

By the definition of the decoder,

$$\Delta(\hat{s}, z) \leq \Delta(s, z).$$

Since the encoder is online, s and $\hat{s}$ agree on their first $i - 1$ output blocks. Subtracting the common contribution of these blocks gives

$$\Delta(\hat{s}_{[i:t]}, z_{[i:t]}) \leq \Delta(s_{[i:t]}, z_{[i:t]}),$$

where for every vector u , we denote by $u_{[i:t]} = (u_i, \dots, u_t)$. By the distance property of $\tilde{T}$,

$$\begin{aligned} \alpha L(t - i + 1) &\leq \Delta(s_{[i:t]}, \hat{s}_{[i:t]}) \\ &\leq \Delta(s_{[i:t]}, z_{[i:t]}) + \Delta(z_{[i:t]}, \hat{s}_{[i:t]}) \\ &\leq 2\Delta(s_{[i:t]}, z_{[i:t]}) \\ &= 2C_P([i, t]). \end{aligned}$$

Thus every $t \in \mathcal{B}_P$ has a witness interval $I_t = [i, t]$ satisfying

$$C_P(I_t) \geq \frac{\alpha L}{2} |I_t|. \quad (4)$$

We select disjoint witness intervals that cover $\mathcal{B}_P$ as follows. Initially, all times in $\mathcal{B}_P$ are unmarked. Choose the largest unmarked time t , select its witness interval $I_t = [i, t]$, and mark every time in $\mathcal{B}_P \cap I_t$. Repeat until all times in $\mathcal{B}_P$ are marked.

After selecting $I_t = [i, t]$, every remaining unmarked time is strictly smaller than i . Hence every subsequently selected interval ends before i . It follows that the selected intervals are pairwise disjoint. Let $\mathcal{I}_P$ denote this collection. Since these intervals cover $\mathcal{B}_P$, we have

$$|\mathcal{B}_P| \leq \sum_{I \in \mathcal{I}_P} |I| \leq \frac{2}{\alpha L} \sum_{I \in \mathcal{I}_P} C_P(I) \leq \frac{2C_P}{\alpha L},$$

where the second inequality follows from (4), and the last inequality uses disjointness.

Since a round is bad if at least one history is decoded incorrectly, and $C = C_A + C_B$, we conclude that

$$\sum_{t=1}^N B_t = |\mathcal{B}_A \cup \mathcal{B}_B| \leq |\mathcal{B}_A| + |\mathcal{B}_B| \leq \frac{2C}{\alpha L}.$$

This proves (3). □

References

- [1] Leonard J. Schulman. Deterministic coding for interactive communication. In *Proceedings of the 25th Annual ACM Symposium on Theory of Computing (STOC)*, pages 747–756. ACM, 1993.